Regression Analysis

Introduction

Prediction or estimation is one of the major problems in almost all spheres of human activity. The estimation or prediction of future production, consumption, prices, investments, sales, profits, income etc., are of paramount importance to a businessman or an economist. Regression analysis is one of the very scientific techniques for making such predictions.

Definition

Regression analysis, in the general sense, means the estimation or the prediction of the unknown value of one variable from the known value of the other variable. It is a statistical device used to study the relationship between two or more variables that are related.

In the words of M. M. Blair " Regression analysis is a mathematical measure of the average relationship between two or more variables in terms of the original units of the data "

Dependent and Independent Variables

In regression analysis there are two types of variables. The variable whose value is influenced or is to be predicted is called dependent variable and the variable which influences the values or is used for prediction, is called independent variable.

Simple and Multiple regressions

On the basis of number of variables, regression can be classified into Simple, and Multiple Regressions.

When there are only two variables the regression equation obtained is called simple regression equation.

In multiple regression analysis there are more than two variables and we try to find out the effect of two or more independent variables on one dependent variable. Let X, Y and Z be three variables. Let X and Y be the independent variables and Z be depending on them. Then we use multiple regression analysis to study the relative movement of Z, for a unit movement in X and Y.

For example: If there are three variables yield, rainfall and temperature. Then suppose yield is depending on rainfall and temperature then we get the regression equation of Z on X and Y where Z is yield, X- rainfall and Y - temperature.

A multiple regression equation is an equation for estimating the value of a depending variable say Z, from values of the independent variables X and Y and it is called a regression equation of Z on X and Y.

Linear and Non-Linear Regression

On the basis of proportion of changes in the variables, the regression can be classified into Linear and Non - linear regressions.

If the given bivariate data are plotted on a graph, the points so obtained on the scatter diagram will more or less concentrate around a curve, called 'curve of regression'.

If the regression curve is a straight line, we say that there is linear regression between the variables under study. The equation of such a curve is the first degree equation in the variables x and y. Mathematically, the relation between x and y in a linear regression, can be expressed in the form, $y = a + bx$. In a linear regression, the change in the dependent variable is proportionate to the changes in the independent variable. If the curve of regression is not a straight line, then the regression is termed as curved or non - linear regression. The regression equation in such cases is not of first degree. In this case the dependent variable does not change by a constant amount of change in the independent variable.

Line of best fit (Regression line)

When the given bivariate data are plotted on a graph, we get the scatter diagram. If the points of the scatter diagram concentrate around a straight line, that line is called the line of best fit. That is, the line of best fit is that line which is closer to the points of the scatter diagram. This line is also known as Regression line. So a regression line is a graphic technique to show the functional relationship between the dependent and the independent variables. It shows average relationship between the variables.

Method of drawing regression lines

The regression lines can be drawn by the two methods (1) Free hand curve method (2) The method of least squares.

Freehand Curve method

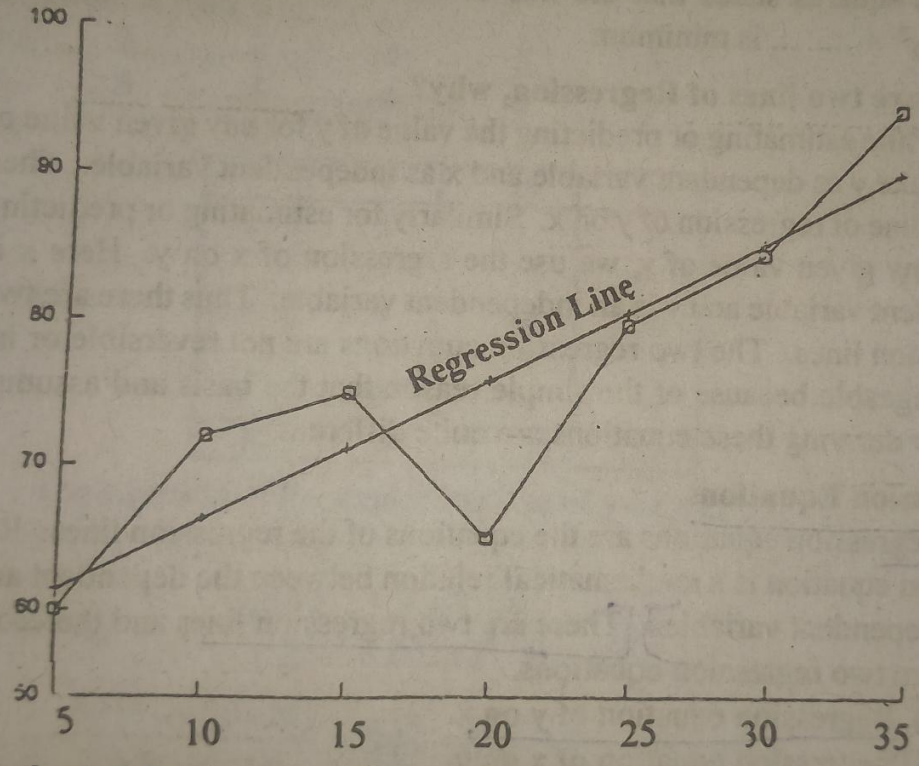
A freehand curve method is an easy method for obtaining a regression line. According to this method original data are plotted on a graph paper. Usually original data when plotted on a graph gives a wave like curve but it depicts a general tendency of the data. Independent variable is taken along the horizontal axis and dependent variable along the vertical axis. We draw a smooth freehand line in such a way that it clearly indicates the tendency of the original data. This line is fitted merely by

inspection. There is no formula or statistical criterion whereby the adequacy of such a line can be judged and it all depends on the judgement of the statistician. Care is taken that the line and the freehand curve be drawn in such a way that the area of the curve below and above the line are approximately equal. Such a line will describe the general tendency of the original data. This is the regression line.

The following are the price in thousands of Rupees and Corresponding supply.

Price	:	5	10	15	20	25	30	35
Supply (in '000 rupees)	:	60	72	75	65	80	85	95

The regression line of supply depending on price by freehand curve method is shown below.

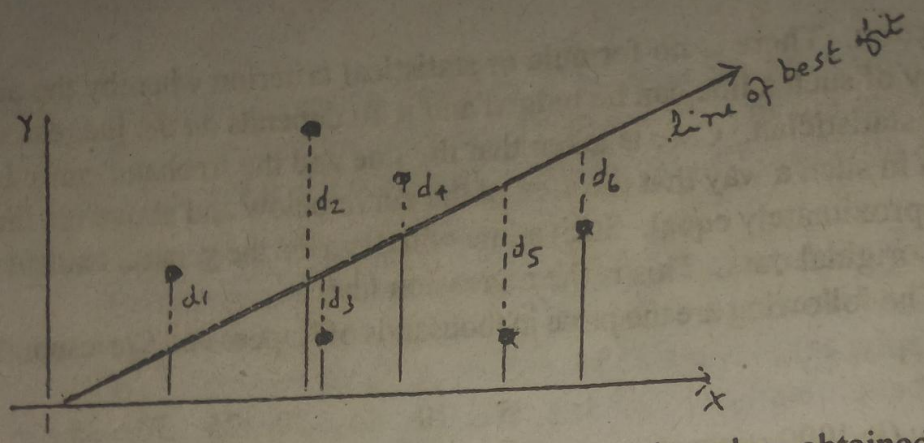


Method of least squares (Curve fitting)

Method of least squares is a method of drawing regression line by applying principle of least squares.

The principle of least squares is that principle which states that the line of best fit should be drawn in such a manner that the sum of the squares of difference between the known values of the dependent variable and the corresponding values of it obtained from the line of best fit should be the least.

ie, $\Sigma (y_0 - y_e)^2$ should be the least where y_0 stands for known (or given) values of dependent variable and y_e stands for the corresponding value of the dependent variable obtained from the line.



The difference between given y values and y values obtained from the line of best fit are respectively d_1, d_2, ... Therefore principle of least squares states that the line of best fit be so drawn such that $d_1^2 + d_2^2 + ...$ is minimum.

There are two lines of Regression, why?

While estimating or predicting the value of y for any given value of x, we take y as dependent variable and x as independent variable. Then we get line of regression of y on x. Similarly for estimating or predicting x for any given value of y, we use the regression of x on y. Here x is dependent variable and y is an independent variable. Thus there are two regression lines. The two regression equations are not reversible or interchangeable because of the simple reason that the basis and assumption for deriving these equations are quite different.

Regression Equations

Regression equations are the equations of the regression lines. Regression equation is a mathematical relation between the dependent and the independent variables. There are two regression lines and therefore there are two regression equations.

(i) Regression equation of y on x.

(ii) Regression equation of x on y.

Derivation of regression equation of y on x when the relation is linear.

Ans: Let $y = a + bx$ be the equation of the regression line of y on x.

To find the best value of a and b apply the method of least squares. According to the method of least squares, a and b should be so chosen that $\Sigma [y - (a + bx)]^2$ is minimum. Here 'y' representing observed values of y and $a + bx$ representing the values of y obtained from the equation. Applying the method of calculus we can minimise $\Sigma [y - (a + bx)]^2$. Consequently, we get the following equations called normal equations.

$\Sigma y = na + b\Sigma x$

$\Sigma xy = a\Sigma x + b\Sigma x^2$

Solving the two equations, we get, $b = \dfrac{n\Sigma xy - (\Sigma x \times \Sigma y)}{n\Sigma x^2 - (\Sigma x)^2}$

From first equation, we get

$$a = \frac{\Sigma y - b\Sigma x}{n} = \frac{\Sigma y}{n} - b\frac{\Sigma x}{n} = \bar{y} - b\bar{x}$$

$\therefore$ The regression equation of y on x becomes

$y = a + bx = (\bar{y} - b\bar{x}) + bx$

$\therefore\ y - \bar{y} = b(x - \bar{x})$

Read b as b_{yx}.

which is also called regression coefficient of y on x.

$\therefore$ The equation becomes, $y - \bar{y} = b_{yx}(x - \bar{x})$

Ex. 1: From the following data of values of x and y, find the regression equation of y on x.

x	:	2	3	4	5	6
y	:	3	5	4	8	9

Ans :

x	y	xy	x^2
2	3	6	4
3	5	15	9
4	4	16	16
5	8	40	25
6	9	54	36
20	29	131	90

The equation of the regression line of y on x is

$y - \bar{y} = b_{yx}(x - \bar{x})$

where $b_{yx} = \dfrac{n\Sigma xy - (\Sigma x \times \Sigma y)}{n\Sigma x^2 - (\Sigma x)^2}$

$$= \frac{5\times131-(20\times29)}{5\times90-(20)^2} = \frac{655-580}{450-400} = \frac{75}{50} = 1.50$$

$$\bar{x} = \frac{\Sigma x}{n} = \frac{20}{5} = 4 \text{ and } \bar{y} = \frac{\Sigma y}{n} = \frac{29}{5} = 5.8$$

The equation is

$y - 5.8 = 1.5(x - 4)$

$y - 5.8 = 1.5x - 6$

$y = 1.5x - 6 + 5.8$

$\underline{y = 1.5x - .2}$.

Derivation of regression of x on y when the relation is linear.

Ans: Assume the equation to be in the form $x = a + by$.

To find the best values of a and b apply the method of least squares. As per the principle of least squares, a and b should be so chosen that $\Sigma[x - (a + by)]^2$ is minimum. Here 'x' representing observed values of x

and a + by representing the values of x obtained from the equation $x = a + by$. Applying the method of calculus we can minimise $\Sigma[x-(a+by)]^2$. Then we get the following normal equations.

$$\Sigma x = na + b\Sigma y$$
$$\Sigma xy = a\Sigma y + b\Sigma y^2$$

Solving these two equations, we get the values of a and b.

ie $$b = \frac{n\Sigma xy - (\Sigma x \times \Sigma y)}{n\Sigma y^2 - (\Sigma y)^2}$$

From the first equation, we get the value of a

$$a = \frac{\Sigma x - b\Sigma y}{n} = \frac{\Sigma x}{n} - b\frac{\Sigma y}{n} = \bar{x} - b\bar{y}$$

Substituting the value of a and b in the equation $x = a + by$

$$x = \bar{x} - b\bar{y} + by$$
$$\therefore \quad x - \bar{x} = b(y - \bar{y})$$

Read b as b_{xy} which is also called regression coefficient of x on y.

The equation becomes $x - \bar{x} = b_{xy}(y - \bar{y})$

Ex. 2: From the data given below find the regression equation of x on y

x	:	5	6	7	3	2
y	:	4	5	8	2	1

Ans:

x	y	xy	y^2
5	4	20	16
6	5	30	25
7	8	56	64
3	2	6	4
2	1	2	1
23	20	114	110

The equation of regression line of x on y is $x - \bar{x} = b_{xy}(y - \bar{y})$

where $$b_{xy} = \frac{n\Sigma xy - (\Sigma x \times \Sigma y)}{n\Sigma y^2 - (\Sigma y)^2}$$

$$= \frac{5 \times 114 - (231 \times 20)}{5 \times 110 - (20)^2} = \frac{570 - 460}{550 - 400} = \frac{110}{150} = 0.73$$

$$\bar{x} = \frac{\Sigma x}{n} = \frac{23}{5} = 4.6 \text{ and } \bar{y} = \frac{\Sigma y}{n} = \frac{20}{5} = 4$$

The regression equation of x on y is $x - 4.6 = .73(y - 4)$

$x = .73 - 2.92 + 4.6$

$$x = .73y + 1.68$$

Summary

Regression equation of y on x is $y - \bar{y} = b_{yx}(x - \bar{x})$

Regression equation of x on y is $x - \bar{x} = b_{xy}(y - \bar{y})$

where $b_{yx} = \dfrac{n\sum xy - (\sum x \times \sum y)}{n\sum x^2 - (\sum x)^2}$ and $b_{xy} = \dfrac{n\sum xy - (\sum x \times \sum y)}{n\sum y^2 - (\sum y)^2}$

b_{yx} and b_{xy} are known as regression coefficients.

Note: (I) For a given value of x, to estimate the value of y we use regression equation of y on x.

(ii) To estimate the value of x for a given value of y, we use the regression equation of x on y.

Ex. 3: From the following data of the age of Husband and the age of wife, form the two regression equations and calculate the husband's age when the wife's age is 16.

Husband's age:	**36**	**23**	**27**	**28**	**28**	**29**	**30**	**31**	**33**	**35**
Wife's age:	**29**	**18**	**20**	**22**	**27**	**21**	**29**	**27**	**29**	**28**

Also find the age of wife when husband's age is 40.

Ans: Take Husband's age as x and wife's age as y.

x	y	xy	x^2	y^2
36	29	1044	1296	841
23	18	414	529	324
27	20	540	729	400
28	22	616	784	484
28	27	756	784	729
29	21	609	841	441
30	29	870	900	841
31	27	837	961	729
33	29	957	1089	841
35	28	980	1225	784
300	250	7623	9138	6414

$\bar{x} = \dfrac{\sum x}{n} = \dfrac{300}{10} = 30$ and $\bar{y} = \dfrac{\sum y}{n} = \dfrac{250}{10} = 25$

$$b_{yx} = \frac{n\sum xy - (\sum x \times \sum y)}{n\sum x^2 - (\sum x)^2} = \frac{10 \times 7623 - (300 \times 250)}{10 \times 9138 - (300)^2}$$

$$= \frac{76230 - 75000}{91380 - 90000} = \frac{1230}{1380} = 0.89$$

$$b_{xy} = \frac{n\sum xy - (\sum x \times \sum y)}{n\sum y^2 - (\sum y)^2} = \frac{10 \times 7623 - (300 \times 250)}{10 \times 6414 - (250)^2} =$$

$$\frac{76230 - 75000}{64140 - 62500} = \frac{1230}{1640} = 0.75$$

Equation of the line of regression of y on x

$y - \bar{y} = b_{yx}(x - \bar{x})$

$y - 25 = 0.89(x - 30)$

$y = 0.89x - 26.7 + 25$

$y = .89x - 1.7$

Equation of the line of regression of x on y

$x - \bar{x} = b_{yx}(y - \bar{y})$

$x - 30 = 0.75(y - 25)$

$x = 0.75y - 18.75 + 30$

$x = .75y + 11.25$

i) When wife's age (y) is 16 Husband's age (x) is obtained by putting y = 16 in the equation of 'x' on 'y'.

$x = 0.75(16) + 11.25 = 23.25$

Husband's age = 23.25

(ii) When the Husband's age (x) is 40, wife's age (y) is obtained by putting x = 40 in the equation of y on x, which is equal to .89 (40) – 1.7 = 35.6 – 1.7 = 33.9

Ex. 4: For 17 observations on price (x) and supply (y), the following data were obtained in appropriate units. $\Sigma x = 544$, $\Sigma x^2 = 19040$, $\Sigma y = 244$, $\Sigma y^2 = 3773$, $\Sigma xy = 8413$, obtain the equations of the two regression lines. What is the supply when price is Rs. 35?

Ans:

$$\bar{x} = \frac{\Sigma x}{n} = \frac{544}{17} = 32; \quad \bar{y} = \frac{\Sigma y}{n} = \frac{244}{17} = 14.35.$$

$$b_{yx} = \frac{n\Sigma xy - (\Sigma x \times \Sigma y)}{n\Sigma x^2 - (\Sigma x)^2} = \frac{17 \times 8413 - (544 \times 244)}{17 \times 19040 - (544)^2} = .37$$

$$b_{xy} = \frac{n\Sigma xy - (\Sigma x \times \Sigma y)}{n\Sigma y^2 - (\Sigma y)^2} = \frac{17 \times 8413 - (544 \times 244)}{17 \times 3773 - (244)^2} = 2.23$$

The regression equation of y on x is $y - \bar{y} = b_{yx}(x - \bar{x})$

ie., $y - 14.35 = .37(x - 32)$

$\therefore \quad y = .37x - 11.84 + 14.35$

ie., $y = .37x + 2.51$

The regression equation of x on y is $x - \bar{x} = b_{xy}(y - \bar{y})$

$x - 32 = 2.23(y - 14.35)$

$x = 2.23y - 1.44$

Given, Price (x) = 35

$\therefore$ supply (y) = .37x + 2.51 = .37 (35) + 2.51 = 15.46

Change of origin does not affect the regression coefficient. So we can replace x by d_x and y by d_y.

$$\therefore\ b_{yx}=\frac{n\Sigma d_x d_y-(\Sigma d_x\times\Sigma d_y)}{n\Sigma d_x^2-(\Sigma d_x)^2}\quad\text{and}\quad b_{xy}=\frac{n\Sigma d_x d_y-(\Sigma d_x\times\Sigma d_y)}{n\Sigma d_y^2-(\Sigma d_y)^2}$$

Ex. 5: A pannel of two judges P and Q graded seven dramatic performances by independently awarding marks as follows:

Performance	:	1	2	3	4	5	6	7
Marks by P	:	46	42	44	40	43	41	45
Marks by Q	:	40	38	36	35	39	37	41

The eighth performance, for which judge Q could not attend, was awarded 37 marks by judge P. If judge Q has also been present, how many marks would be expected to have been awarded by him to the eighth performance.

Ans: Let x stand for marks by P and y stands for marks by Q.

x	y	d_x = x − 42	d_y = y − 39	d_x^2	d_y^2	$d_x d_y$
46	40	4	1	16	1	4
42	38	0	−1	0	1	0
44	36	2	−3	4	9	−6
40	35	-2	−4	4	16	8
43	39	1	0	1	0	0
41	37	−1	−2	1	4	2
45	41	3	2	9	4	6
301	266	7	−7	35	35	14

Here only one regression equation is required. ie y on x

$$\bar{x}=\frac{\Sigma x}{n}=\frac{301}{7}=43\ \text{and}\ \bar{y}=\frac{\Sigma y}{n}=\frac{266}{7}=38$$

$$b_{yx}=\frac{n\Sigma d_x d_y-(\Sigma d_x\times\Sigma d_y)}{n\Sigma d_x^2-(\Sigma d_x)^2}=\frac{7\times14-(7\times-7)}{7\times35-(7)^2}=\frac{147}{196}=0.75$$

The equation of the line of regression of y on x is

$y-\bar{y}=b_{yx}(x-\bar{x})$

$y-38=0.75\,(x-43)=0.75x-32.25$

$\therefore\ y=0.75x-32.25+38$

$y=0.75x+5.75$

When x = 37, the estimate of y is given by

$y=(0.75\times37)+5.75=27.75+5.75=33.50$

Hence if the judge Q were present at the eighth performance he would have assigned the score 33.5

When deviations are taken from actual mean so that $d_x = x - \bar{x}$ and $d_y = y - \bar{y}$ then $\Sigma d_x = 0$ and $\Sigma d_y = 0$

Then $b_{yx} = \frac{\Sigma dx\, dy}{\Sigma d_x^2}$ and $b_{xy} = \frac{\Sigma dx\, dy}{\Sigma dy^2}$

Ex. 6: From the following data, obtain the two regression equations.

Sales	**:**	**91**	**97**	**108**	**121**	**67**	**124**	**51**	**73**	**111**	**57**
Purchases:		**71**	**75**	**69**	**97**	**70**	**91**	**39**	**61**	**80**	**47**

Ans: Let us denote the sales by the variable x and the purchase by the variable y. Let us take deviations from actual mean.

$\bar{x} = \frac{\Sigma x}{n} = \frac{900}{10} = 90$ and $\bar{y} = \frac{\Sigma y}{n} = \frac{700}{10} = 70$

x	y	$d_x = x - \bar{x}$	$d_y = y - \bar{y}$	d_x^2	d_y^2	$d_x d_y$
91	71	1	1	1	1	1
97	75	7	5	49	25	35
108	69	18	–1	324	1	–18
121	97	31	27	961	729	837
67	70	–23	0	529	0	0
124	91	34	21	1156	441	714
51	39	–39	–31	1521	961	1209
73	61	–17	–9	289	81	153
111	80	21	10	441	100	210
57	47	–33	–23	1089	529	759
900	700	0	0	6360	2868	3900

$b_{yx} = \frac{\Sigma d_x d_y}{\Sigma d_x^2} = \frac{3900}{6360} = .61$

$b_{xy} = \frac{\Sigma d_x d_y}{\Sigma d_y^2} = \frac{3900}{2868} = 1.36$

Equation of line of regression of y on x is $y - \bar{y} = b_{yx}(x - \bar{x})$

$\therefore\ y - 70 = 0.61\,(x - 90) = 0.61x - 54.9$

$\therefore\ y = 0.61x - 54.9 + 70.00$

$\therefore\ \underline{y = 0.61x + 15.1}$

Equation of line of regression of x on y is $x - \bar{x} = b_{xy}(y - \bar{y})$

$\therefore\ x - 90 = 1.36\,(y - 70) = 1.36 - 95.2$

$\therefore\ x = 1.36 - 95.2 + 90.00$

$\therefore\ \underline{x = 1.36x - 5.2}$

Relation between Correlation Coefficient and Regression Coefficients

$$b_{yx} = r\frac{\sigma_y}{\sigma_x} \text{ and } b_{xy} = r\frac{\sigma_x}{\sigma_y}$$

Ex. 7: You are given the following data:

	x	y
Arithmetic Mean	**36**	**85**
Standard deviation	**11**	**8**

Correlation coefficient between x and y = 0.66

(i) Find the two regression equations

(ii) Estimate the value of x when y = 75

Ans:

Given $\bar{x} = 36$, $\bar{y} = 85$, $\sigma_x = 11$, $\sigma_y = 8$, $r = 0.66$

$$b_{yx} = r\frac{\sigma_y}{\sigma_x} = \frac{0.66 \times 8}{11} = .48$$

$$b_{xy} = r\frac{\sigma_x}{\sigma_y} = \frac{0.66 \times 11}{8} = .91$$

Equation of regression of y on x

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

$$\therefore \quad y - 85 = 0.48(x - 36) = 0.48x - 17.28$$

$$\therefore \quad \underline{y = 0.48x + 67.72}$$

Equation of regression of x on y

$$x - \bar{x} = b_{xy}(y - \bar{y})$$

$$\therefore \quad x - 36 = 0.91(y - 85) = 0.91y - 77.35$$

$$\therefore \quad \underline{x = 0.91y - 41.35}$$

To estimate the value of x when y = 75 we use the regression equation of x on y.

$$x = .91(75) - 41.35 = 68.25 - 41.35 = \underline{26.9}$$

Ex. 8: From the following results, estimate the yield of crops when the rainfall is 22 cms and the rainfall when the yield is 600 kgs.

	Yield in Kgs (Y)	Rainfall in cms (X)
Mean	**508.4**	**26.7**
S.D.	**36.8**	**4.6**

Coefficient of correlation between yield and rainfall is 0.52

Ans: Given $\bar{x} = 26.7$, $\bar{y} = 508.4$

$\sigma_x = 4.6$, $\sigma_y = 36.8$, $r = 0.52$

Equation of line of regression of y on x is $y - \bar{y} = b_{yx}(x - \bar{x})$

where $b_{yx} = r\frac{\sigma_y}{\sigma_x} = \frac{.52 \times 36.8}{4.6} = 4.16$

$y - 508.4 = 4.16(x - 26.7) = 4.16x - 111.072$

$y = 4.16x + 397.328$

The most likely yield for crop (y) when the rainfall (x) = 22 cms is given by $\underline{y = (4.16 \times 22) + 397.328 = 91.52 + 397.328 = 488.848 \text{kgs.}}$

Equation of line of regression of x on y is $x - \bar{x} = b_{xy}(y - \bar{y})$

where $b_{xy} = r\frac{\sigma_x}{\sigma_y} = \frac{.52 \times 4.6}{36.8} = 0.065$

$x - 26.7 = 0.065(y - 508.4)$

$x = 0.065y - 33.046 + 26.7$

$x = 0.065y - 6.346$

The most likely estimate of rainfall (x) corresponding to the yield y = 600 (kgs) is given by

$\underline{x = 0.065 \times 600 - 6.346 = 39.00 - 6.346 = 32.654 \text{cms.}}$

How to get b_{xy} and b_{yx} from Regression equations ?

Ans: b_{yx} is obtained from the equation of y on x and b_{xy} from x on y.

When the regression equation of y on x is expressed in the form $y = ax + b$, then 'a' is b_{yx}. Similarly when the regression equation of x on y is expressed in the form $x = cy + d$, 'c' is b_{xy}.

Ex. 9: Find b_{yx} if $2x + 4y - 5 = 0$ is the equation of y on x

Ans: $2x + 4y - 5 = 0$ $\quad \therefore \; 4y = -2x + 5$

or $y = -.5x + 1.25$ $\quad$ Here $a = -.5$ $\quad \therefore \; b_{yx} = -.5$

Ex. 10: Find b_{xy} if $3x + 2y + 4 = 0$ is the equation of x on y.

$3x + 2y + 4 = 0$ $\quad \therefore \; 3x = -2y - 4$

$x = -.67y - 1.33$ $\quad$ Here $c = -.67$ $\quad \therefore \; b_{xy} = -.67$

How to get $\bar{x}$ and $\bar{y}$ from the regression equations ?

Ans: Solve the two regression equations. The values of x and y obtained are means of x and y.

Ex. 11: Find the mean of variables x and y, given the following

Regression of y on x : $2y - x - 50 = 0$

Regression of x on y : $3y - 2x - 10 = 0$

Ans: The means of the variables x and y are obtained by solving the two regression eqautions.

Multiply First equation by 2 $\quad 4y - 2x = 100$

Second equation is $\quad 3y - 2x = 10$

Subtracting, $\quad y = 90$

Substituting in the first equation

$2 \times 90 - x - 50 = 0$

$x = 2 \times 90 - 50 = 130$

Hence means are given by : $\underline{\bar{x} = 130, \; \bar{y} = 90}$

How to identify the two regression equations ?

Ans: By supposing one of the equations as the regression of y on x and the other as x on y, we can obtain b_{yx} and b_{xy}. If the product of these two is numerically not greater than 1 then our supposition is true. However if $b_{yx} \times b_{xy}$ numerically is greater than 1, then our supposition is wrong. Then we shall conclude in the other way.

Ex. 12: Out of the two lines of regression given by
$x + 2y - 5 = 0$ and $2x + 3y - 8 = 0$
Which one is the regression line of x on y ?

Ans: $x + 2y - 5 = 0$ ..... (i)

$2x + 3y - 8 = 0$ ..... (ii)

Let us assume the equation (i) as the regression of x on y and equation (ii) as the regression of y on x.

Write the equation (i) to have x in L H S.

$x = -2y + 5$. $\therefore b_{xy} = -2$

Write the equation (ii) to have y in L H S.

$y = -\frac{2}{3}x + \frac{8}{3}$ $\therefore b_{xy} = -\frac{2}{3}$

$b_{xy} \times b_{yx} = (-2) \times (-\frac{2}{3}) = \frac{4}{3} = 1.33$ which is > 1.

$\therefore$ Our supposition is wrong. Hence it should be in the other way.

$\therefore$ (i) is the line of regression of y on x and (ii) is the line of regression of x on y.

NOTE: $b_{yx} \times b_{xy} = r\frac{\sigma_y}{\sigma_x} \times r\frac{\sigma_x}{\sigma_y} = r^2$ $\boxed{r = \sqrt{b_{yx} \times b_{xy}}}$

When r is positive both b_{yx} and b_{xy} are positive. If r is negative, both b_{yx} and b_{xy} are negative. So b_{yx} and b_{xy} will not be opposite sign.

Ex. 13: Find 'r' if $b_{yx} = -.2$ and $b_{xy} = -.7$

Ans: $r = \sqrt{b_{yx} \times b_{xy}} = \sqrt{-.2 \times -.7} = \sqrt{.14} = \underline{-.37}$

Ex. 14: If $b_{yx} = .83$, $\sigma_x = 20$, $\sigma_y = 12$, find r.

Ans: We know $b_{yx} = r\frac{\sigma_y}{\sigma_x}$

$.83 = r\frac{12}{10}$

$.83 = r \times 1.2$

$\therefore r = \frac{.83}{1.2} = .7$

Ex. 15: In the study of regression equations, following values were obtained. Regression coefficient of y on x = .25, r = .42, S. D of y = 4, find S. D of x.

Ans: Given $b_{yx} = .25$, $r = .42$, $\sigma_y = 4$.

We know that $b_{yx} = r\frac{\sigma_y}{\sigma_x}$

$\therefore .25 = \frac{.42 \times 4}{\sigma_x}$ $\therefore .25\sigma_x = 1.68$

$\sigma_x = \frac{1.68}{.25} = \underline{6.72}$

Ex. 16: In a partially destroyed record of an analysis of correlation data the following results only are legible.

Variance of x = 9

Regression equations:

$8x - 10y + 66 = 0$

$40x - 18y = 214$

Find: (i) The mean values of X and Y (ii) The co-efficient of correlation (iii) The standard deviation of Y.

Ans: $8x - 10y = -66$

$40x - 18y = 214$

Multiplying equation (i) by 5 $\quad 40x - 50y = -330$

(ii) is subtracting $\quad 40x - 18y = 214$

$-32y = -554$

$\therefore \quad y = 17$

Putting the value of y in equation (i) $8x - (10 \times 17) = -66$

$\therefore \quad x = 13$

Thus $\underline{\bar{y} = 17 \text{ and } \bar{x} = 13}$

(ii) Taking first equation to be equation of x on y

$8x = 10y - 66$

$\therefore \quad x = \frac{10}{8} y - \frac{66}{8}$

Take second equation as y on x.

$-18y = -40x + 214$

$\therefore \quad y = \frac{40}{18} x - \frac{214}{18}$

The product of coefficients $\frac{10}{8} \times \frac{40}{18} = \frac{400}{144}$ which exceeds 1. $\therefore$ our assumption is wrong. Hence the first equation is of y on x and second is x on y.

Write the first equation as

$-10y = -8x - 66$

$y = \frac{8}{10} x + \frac{66}{10} \qquad \therefore b_{yx} = \frac{8}{10} = .8$

Equation (ii) is

$40x = 18y + 214$

$x = \frac{18}{40} y + \frac{214}{40} \qquad \therefore b_{xy} = \frac{18}{40} = .45$

$r = \sqrt{b_{yx} \times b_{xy}} = \sqrt{.8 \times .45} = \sqrt{.36} = \underline{0.6}$

(ii) Given variance of x = 9. $\qquad \therefore \sigma_x = \sqrt{9} = 3$

We have $b_{yx} = r \frac{\sigma_y}{\sigma_x}$

Putting $r = .6$, $\sigma_x = 3$, and $b_{yx} = .8$ in the above relation

$.8 = .6 \frac{\sigma_y}{3}$

$$\therefore \sigma_y = \frac{.8 \times .3}{.6} = \frac{4}{=}$$

Ex. 17: Comment on the following results.
For a bivariate distribution,

1) **Coefficient of regression of y on x is 4.2 and Coefficient of regression of x on y is 0.50**
2) **$b_{xy} = -.82$ and $b_{yx} = .25$**

Ans: (1) $r^2 = b_{yx} \times b_{xy} = 4.2 \times 0.5 = 2.10$

But r^2 cannot be greater than 1. Hence the given statement is wrong.

2) b_{yx} and b_{xy} can not be opposite sign. So the statement is wrong.

Usefulness of the study of regression.

The study of regression is very useful in many types of analysis. By its study, we are able to obtain most probable values of one series for given values of the other related series. If we know that two series relating to price and supply are correlated we can find out what would be the effect on price if the supply of commodity were increased or decreased to a particular level. The utility of regression is very great in physical sciences where the data are generally in functional relationship. Therefore, it is always possible to exactly calculate the value of one variable for a given value of the other variable by studying their regression. Thus the primary use of the regression equation is to describe the relationship between the two variables and to show the rates of change in one factor in terms of another.

Distinction between Correlation and Regression

1. In correlation analysis we study degree of relationship between the variables whereas in regression analysis we study the nature of relationship.
2. In correlation analysis, the choice of dependent and independent variables is purely a personal choice and is of no practical significance. In regression analysis, one has to decide which variable shall be taken as dependent and which as independent.
3. Correlation analysis is not for the purpose of prediction whereas the regression analysis is basically used for prediction purposes.